

Meeting 1015

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October 10, 2025

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- ▶ 樣本數 1567
- ▶ 變數 Time / x1 / x2 / ... / x590 / PassFail
 - Features: 591, Y: PassFail (binary classification)
 - Time: 2008-07-19 11:55:00
 - PassFail: Pass $\rightarrow$ 0, Fail $\rightarrow$ 1 (14:1)
 - Missing values: 41951

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- Data preprocessing
- KNN-imputation
- Random forest features selection
- imbalance - SMOTE to 1:1
- 0.8 training, 0.2 testing by decision tree

● 結果

- Best: ACC 86.94%, FAR 9.22%
- 反查 Fail 節點

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- ▶ K-folds cross validation
 - Robust performance
- ▶ $y = \hat{f}(x)$
 - Important features

Li-Fu Lai (2022)。使用 KNN 和決策樹方法對填補缺失值的製造數據進行失效預測和原因追蹤分析

The End

Questions & Comments